

PROJECT PEAR

Quantitative Valuation & Momentum Analysis

A Major Consumer Technology Company

Analysis Date: 2026-06-18
Data Period: 2021-06-22 to 2026-06-18

1. Executive Summary

This report presents a quantitative valuation and momentum analysis of Pear Corporation, a major consumer technology company. The analysis covers five years of daily price data (2021-06-22 to 2026-06-18) and combines a Monte Carlo DCF model, comparable company analysis, technical momentum indicators, formal event studies, and GARCH volatility modelling.

Key findings from this analysis.

- The Monte Carlo DCF (50,000 simulations) produces a median fair value of \$100.97 per share against a current price of \$297.77. This implies -66.1% upside (or downside). The probability the stock is undervalued on a pure DCF basis is 0.0%.
- Relative valuation shows Pear trading at a premium across all metrics. The P/E z-score is 1.28 standard deviations above the peer mean. The EV/EBITDA z-score is 0.76.
- Momentum is mixed. The long-term trend is bullish (SMA50 above SMA200, 1-year return of 52.8%). But short-term indicators are neutral to bearish. RSI at 38.8 and MACD below its signal line suggest cooling momentum.
- Event studies find two statistically significant events. The Q2 FY2024 earnings plus buyback announcement generated +8.58% CAR ($p=0.028$). The Q1 FY2024 report showed -10.02% CAR ($p=0.005$). Most other events were not statistically significant at the 5% level.
- The Hurst exponent of 0.4456 with a bootstrap 95% CI of [0.368, 0.549] suggests mild mean-reverting behaviour. This is not strong enough to trade on, but it weakens the case for pure momentum strategies.

2. Data and Methodology

2.1 Data Sources

Daily OHLCV price data for Pear Corporation and five technology sector peers was obtained via the Yahoo Finance API. The analysis period spans 2021-06-22 to 2026-06-18, giving 1254 trading day observations. The S&P 500 index serves as the market benchmark for beta estimation and event study market models.

2.2 Methodology Overview

The analysis combines six quantitative frameworks.

- Monte Carlo DCF valuation. 50,000 simulations with stochastic revenue growth (triangular distribution), FCF margin (normal), WACC (normal), and terminal growth (normal). Five-year projection with growth decay toward terminal rate.
- Comparable company analysis. Six large-cap technology peers compared on P/E, Forward P/E, P/B, and EV/EBITDA. Z-scores computed to measure Pear's position within the peer distribution.
- Technical momentum analysis. RSI(14), MACD(12,26,9), Bollinger Bands(20,2), moving average crossovers (50/200), multi-period price momentum, relative strength vs S&P 500, and Hurst exponent for regime characterisation.
- Event study methodology. Market model estimated on 252-day pre-event window. Abnormal returns computed over a [-5, +10] day event window. Cumulative abnormal returns (CAR) tested for significance via t-test.
- GARCH(1,1)-t volatility modelling. Conditional volatility estimated with Student-t innovations. Rolling 6-month beta against the S&P 500.
- Statistical robustness. Block bootstrap (block size=20, 1000 iterations) for confidence intervals on key parameters. CUSUM test for structural breaks in the return series.

2.3 DCF Parameter Justification

The WACC is derived from the CAPM. The risk-free rate of 4.3% reflects current US 10-year Treasury yields. The equity risk premium of 5.5% follows the Damodaran estimate for the US market. Beta of 1.086 is estimated from regression of Pear returns on S&P 500 returns.

Revenue growth uses a triangular distribution with parameters (2%, 6%, 12%). The base case of 6% aligns with the company's recent historical revenue growth of 6.4%. Growth decays toward the terminal rate over the projection period. This decay prevents compounding unrealistic growth rates into perpetuity.

The FCF margin of 23.7% is derived from the most recent annual free cash flow divided by revenue. The normal distribution with 3% standard deviation allows for margin expansion or compression.

3. Valuation Analysis

3.1 Monte Carlo DCF Results

The Monte Carlo DCF produces a median fair value of \$100.97 per share (mean \$103.49). The 90% confidence interval runs from \$72.20 to \$142.81. Against the current price of \$297.77, this suggests -66.1% implied return to median fair value.

Only 0.0% of simulated fair values exceed the current price. On a pure DCF basis, the stock appears significantly overvalued. But this finding demands careful interpretation (see Section 3.3).

Sensitivity analysis shows fair value is most sensitive to WACC and revenue growth assumptions. A 1% change in WACC shifts the median fair value by roughly 15-20%. Revenue growth and FCF margin have smaller but material effects.

3.2 Relative Valuation

Pear trades at a premium to all peers across every metric examined.

Metric	Pear	Peer Mean	Z-Score	Percentile
trailing pe	36.05	28.56	1.28	90.0%
forward pe	31.03	22.21	1.50	93.3%
price to book	41.02	15.86	1.73	95.8%
ev to ebitda	27.57	22.20	0.76	77.7%

The P/B ratio stands out at 1.73 standard deviations above the peer mean. This reflects the company's capital-light business model and aggressive share buyback programme, which has reduced book equity substantially. The EV/EBITDA premium (z-score 0.76) is the most moderate, suggesting the market recognises the earnings quality but prices it less aggressively on an enterprise basis.

3.3 The Valuation Gap, Why DCF and Market Disagree

The DCF implies roughly 66% downside. The market obviously disagrees. This is not an error in the model. It is the model honestly showing what a standard DCF produces for a company trading at 36x earnings. Several factors explain the gap.

- **Optionality.** The DCF discounts known cash flows. It does not price optionality in new product categories, services growth, or AI monetisation. The market is pricing some probability of a step-change in growth that the base DCF does not capture.
- **Quality premium.** Pear generates exceptional free cash flow (\$99B annually), has a loyal installed base, and operates in an oligopoly. Investors pay more per dollar of earnings for predictability.
- **Buyback yield.** The company returns enormous amounts of capital through buybacks. This shrinks the float over time, supporting the share price mechanically.
- **The DCF uses a 5-year explicit projection.** For a company with a multi-decade competitive advantage, a longer projection period would capture more of the economic value, reducing the gap.

The honest conclusion is that a standard DCF framework is not sufficient to justify the current price. Either the market is pricing in growth and optionality that the DCF does not model, or the stock is overvalued. This analysis cannot distinguish between those two explanations with certainty.

4. Momentum Analysis

4.1 Technical Indicators

The technical picture shows a long-term uptrend with short-term consolidation.

Indicator	Value	Signal
SMA 50/200	Golden Cross	BULLISH
RSI (14)	38.8	NEUTRAL
MACD	1.19 vs 3.25	BEARISH
Bollinger %B	0.324	MID-RANGE

The SMA 50 sits above the SMA 200, confirming the long-term uptrend. However, RSI at 38.8 is in neutral territory and MACD has crossed below its signal line. This combination typically indicates momentum is fading without yet turning bearish. The Bollinger %B at 0.324 shows the price sitting in the lower half of its 20-day range.

4.2 Multi-Period Momentum

Period	Return
1 Week	0.72%
1 Month	-0.40%
3 Months	19.72%
6 Months	8.64%
1 Year	52.81%

The momentum profile is positive at longer timeframes (1-year return of 52.8%, 3-month of 19.7%) but flat to slightly negative at shorter ones. The 1-month return of -0.40% suggests the recent rally has stalled. This pattern is consistent with a stock entering a consolidation phase after a strong run.

4.3 Relative Strength

The 63-day relative strength versus the S&P 500 is 1.0540. Values above 1.0 indicate outperformance. The 21-day measure at 0.9761 shows recent slight underperformance. This divergence between the short and medium-term relative strength mirrors the momentum picture: strength over 3 months, weakness over 1 month.

4.4 Hurst Exponent and Regime

The Hurst exponent of 0.4456 falls below 0.5, suggesting mild mean-reverting behaviour. The bootstrap 95% CI is [0.368, 0.549]. This confidence interval spans the 0.5 boundary, which means we cannot reject the random walk hypothesis. The evidence for mean-reversion is suggestive but not statistically conclusive.

Practically, this means trend-following strategies may underperform expectations. The stock tends to reverse some of its moves rather than extending them. This is consistent with a large-cap equity in a well-studied sector where mispricings are quickly arbitrated.

5. Event Study Analysis

5.1 Methodology

Each event study follows the standard market model approach. A 252-day estimation window prior to the event window is used to estimate the market model parameters (alpha and beta from OLS regression of Pear returns on S&P 500 returns). Abnormal returns are computed as the difference between actual returns and predicted returns from the market model. The event window spans 5 days before to 10 days after the announcement.

The cumulative abnormal return (CAR) is tested against the null hypothesis of zero abnormal performance using a t-test. The residual standard deviation from the estimation window provides the standard error estimate.

5.2 Results

Event	Date	CAR	t-stat	p-value	Sig.
AI Platform Announcement (WWDC 2024)	2024-06-10	5.92%	1.44	0.1525	NO
Q1 FY2024 Earnings Report	2024-02-01	-10.02%	-2.83	0.0050	YES
Q2 FY2024 Earnings + Buyback Announ	2024-05-02	8.58%	2.21	0.0279	YES
Q3 FY2024 Earnings Report	2024-08-01	0.87%	0.19	0.8523	NO
Q4 FY2024 Earnings Report	2024-10-31	-3.55%	-0.75	0.4516	NO
Q1 FY2025 Earnings Report	2025-01-30	7.71%	1.55	0.1213	NO
Mixed Reality Headset Launch	2024-01-16	-2.47%	-0.70	0.4841	NO
New Product Line Reveal (Sept 2024)	2024-09-09	-3.82%	-0.82	0.4123	NO
Trade Tariff Escalation Impact	2025-04-09	-4.14%	-0.80	0.4253	NO
Q2 FY2025 Earnings Report	2025-05-01	-9.40%	-1.73	0.0845	NO

Two events show statistical significance at the 5% level.

- Q2 FY2024 Earnings plus Buyback Announcement (May 2024). CAR of +8.58% (t=2.21, p=0.028). The large buyback announcement drove significant positive abnormal returns. This suggests the market valued the capital return commitment above the earnings themselves.
- Q1 FY2024 Earnings Report (February 2024). CAR of -10.02% (t=-2.83, p=0.005). A significant negative reaction, despite the company beating earnings estimates. This likely reflects concerns about forward guidance and the China market.

The AI Platform Announcement (WWDC 2024) produced a +5.92% CAR but was not statistically significant (p=0.153). This is a borderline case. The event window captures the full conference period, and other market-moving news may have contaminated the window. The Q2 FY2025 Earnings Report (May 2025) shows -9.40% CAR with p=0.085, just missing the 5% significance threshold.

5.3 Event Study Limitations

- The market model assumes a linear relationship between Pear returns and market returns. Beta instability (range 0.72 to 1.48 over the sample) means the estimation window beta may not represent the event window beta.

- Event windows may overlap. Two events close in time can contaminate each other's abnormal returns.
- The event dates used are public announcement dates. If information leaked before the announcement, the event window may miss the true market reaction.
- Single-factor market model does not control for sector or size effects. A multi-factor model (Fama-French) would provide cleaner abnormal returns.

6. Volatility and Risk Profile

6.1 GARCH(1,1)-t Model

The GARCH(1,1) model with Student-t innovations produces $\alpha = 0.0673$, $\beta = 0.9088$, giving persistence of 0.9762. The degrees of freedom parameter is 4.79, substantially below the normal approximation (infinity), confirming the heavy-tailed nature of the return distribution.

Current annualised conditional volatility is 26.0%, which places the stock in the MODERATE regime relative to its historical distribution. The median annualised volatility is 25.1%, with the 90th percentile at 35.3%.

6.2 Beta Stability

The rolling 6-month beta against the S&P 500 ranges from 0.723 to 1.478. The current reading is 0.839, below the sample mean of 1.214. The bootstrap 95% CI for beta is [1.108, 1.331].

The beta instability matters for valuation. Our DCF uses a point estimate for WACC derived from a single beta. If the true beta is at the upper end of its range, the WACC increases and the fair value decreases further. If the beta is at the lower end (as it currently is), the stock becomes somewhat less overvalued on a DCF basis, but not enough to close the gap.

6.3 Risk Metrics

Metric	Value
VaR 95% (Daily)	-2.71%
VaR 99% (Daily)	-4.45%
CVaR 95% (Daily)	-3.86%
CVaR 99% (Daily)	-5.60%
Max Drawdown	-33.36%
Sharpe Ratio	0.566
Sortino Ratio	0.826
Calmar Ratio	0.617

The maximum drawdown of -33.4% occurred on 2025-04-08. The Sharpe ratio of 0.566 is moderate, with a bootstrap 95% CI spanning [-0.199, 1.462]. The fact that this CI includes negative values means we cannot statistically confirm the Sharpe ratio is positive. Risk-adjusted returns are positive but unremarkable.

7. Statistical Robustness

7.1 Bootstrap Confidence Intervals

Block bootstrap with block size of 20 trading days and 1,000 iterations provides confidence intervals for key parameters. The block structure preserves the serial dependence in returns that a naive bootstrap would destroy.

Parameter	Point Estimate	Bootstrap Mean	95% CI Lower	95% CI Upper
Sharpe Ratio	0.566	0.613	-0.199	1.462
VaR 99%	-4.45%	-4.43%	-5.18%	-3.73%
Beta	0.839	1.214	1.108	1.331
Hurst Exponent	0.446	0.457	0.368	0.549

The Sharpe ratio CI spanning negative to positive is the most practically important finding. It means five years of data is not enough to confidently say Pear outperformed on a risk-adjusted basis. The beta CI is relatively tight, suggesting the systematic risk estimate is more stable. The Hurst exponent CI spanning 0.5 means we cannot reject the random walk.

7.2 Structural Break Analysis

The CUSUM test for structural breaks in the return series yields a maximum statistic of 0.5741 against a 5% critical value of 1.36. This is below the critical value, indicating no significant structural break at the 5% level.

The maximum deviation point is 2025-04-08, which coincides with the tariff-driven market sell-off. While the CUSUM does not formally detect a break, the rolling beta analysis shows substantial variation across the sample. The return-generating process is not perfectly stationary, even if the CUSUM test lacks power to reject.

7.3 Earnings Momentum

Pear has beaten consensus earnings estimates in 22 of 24 reported quarters (91.7% beat rate). The average earnings surprise is 8.17%.

A consistently high beat rate suggests either genuinely strong execution or systematic analyst conservatism in setting estimates. Both interpretations are positive for the stock. However, the magnitude of surprises matters more than frequency for price impact, and the event study results show that even beats can produce negative CARs when the market is focused on forward guidance.

8. Model Limitations and Caveats

This analysis has material limitations that affect interpretation.

DCF model limitations.

- The model does not price optionality in new business lines, product categories, or market entries. For a company investing heavily in AI and services expansion, this omission may explain a large portion of the DCF/market gap.
- Terminal value accounts for the majority of the DCF output. Small changes in terminal growth rate or WACC produce large swings in valuation. The sensitivity table in the Excel dashboard shows this clearly.
- Share buybacks reduce shares outstanding over time. The model uses current shares outstanding as a static denominator, understating per-share fair value for a company that has reduced its share count by roughly 40% over the past decade.

Momentum analysis limitations.

- Technical indicators are backward-looking by construction. They describe what has happened, not what will happen. The predictive power of any single indicator is weak in an efficient large-cap market.
- The Hurst exponent is a statistical measure of the full sample. It does not capture regime changes within the sample. The stock may have been trending in some periods and mean-reverting in others.

Event study limitations.

- The market model is single-factor. A Fama-French three or five-factor model would better isolate abnormal returns from size and value effects.
- Events are identified manually. There is selection bias in choosing which events to study.
- Confounding events within the event window cannot be fully controlled for.

General limitations.

- This analysis uses daily close prices. It does not capture intraday dynamics, options market signals, or after-hours trading reactions to earnings.
- Beta instability (range 0.72 to 1.48) means any single-point beta used for CAPM is questionable. The WACC, and therefore the entire DCF, inherits this uncertainty.
- Survivorship bias. The company has survived and grown over the analysis period. The analysis of a failed company would look very different.
- All models assume the past is informative about the future. Structural changes in the technology industry, regulatory environment, or macroeconomic conditions could render historical parameters unreliable.

9. Conclusions

Pear Corporation presents a classic case of a high-quality company at a premium valuation. The quantitative evidence leads to three main conclusions.

1. The DCF says overvalued. The market says otherwise.

At 298x earnings and a DCF median fair value 66% below market price, a traditional valuation framework cannot justify the current price. This does not necessarily mean the stock is overvalued. It means the market is pricing in growth, optionality, and quality premiums that a standard 5-year DCF does not capture. An investor must decide whether they believe those premiums are justified.

2. Momentum is positive but fading.

The long-term trend remains intact. The stock is above both key moving averages, the 1-year return exceeds 50%, and 3-month relative strength against the S&P 500 is positive. But short-term signals are cooling. MACD has crossed bearish, RSI is mid-range and falling, and 1-month momentum is flat. This is consistent with a consolidation phase rather than a reversal.

3. Earnings momentum is strong but event impact is mixed.

With a 92% earnings beat rate, execution risk is low. But the event studies show that beats alone do not guarantee positive price reactions. Forward guidance, capital allocation decisions, and macro factors (tariffs) have driven the most significant market reactions. The buyback announcement was the single most positive event, suggesting the market values capital return over marginal earnings beats.